

RegimeCast

Regime-aware AI post-processing of monsoon rainfall forecasts

- Problem Statement ID – 26080
- Problem Statement Title – Regime-Aware AI Post-Processing of Monsoon Rainfall Forecasts
- Theme – Smart Automation
- PS Category – Software
- Team ID-
- Team Name (Registered on portal)



RegimeCast

AI that corrects the weather model's rainfall forecast for the monsoon regime it is in

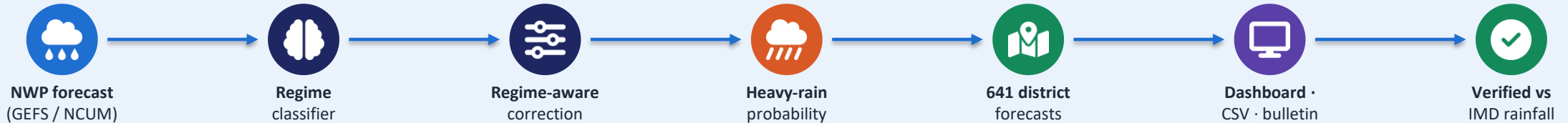
PROBLEM DEFINITION

- ❖ NWP models rain on **too many days** (52% of cell-days vs 37% observed) yet give **too few heavy-rain days** (0.97% vs 1.45%)
- ❖ The errors change with the **weather regime** — active / break spells, monsoon depressions, Western Ghats orography — so one global bias correction fails
- ❖ Disaster managers need **district heavy-rain warnings**, not raw 25 km grid numbers

IDEA / SOLUTION

- ❖ An **AI post-processing layer** on the forecast NCMRWF already makes — it corrects, never replaces, the model
- ❖ Reads the forecast's own **850 hPa wind, pressure & moisture** to identify the regime (Active / Break / Depression / Normal) plus orographic & coastal zones
- ❖ **Regime-aware correction** (LightGBM + U-Net), **heavy-rain probabilities** and **641 district forecasts** with explanations

SYSTEM FLOW



PROBLEM RESOLUTION

- ❖ Day-1 forecast error **-20%** on unseen test years (16.73 → 13.39 mm)
- ❖ Heavy-rain detection **AUC 0.56 → 0.90**; before the Kerala floods, Kottayam got a **61%** heavy-rain chance (base rate 1.4%)
- ❖ **Live mode**: today's NOAA forecast → district product in **< 1 min**, tested on **610 real daily forecasts**

UNIQUE VALUE PROPOSITION

- ❖ **Regime as probability** — the corrector sees the classifier's uncertainty, not a hard guess
- ❖ **Honest by design** — one-time held-out test with locked model hashes; reports where it loses
- ❖ **Deep learning vs trees, head-to-head** — U-Net wins on heavy rain, LightGBM on bias
- ❖ **Offline, open data, laptop CPU**, English + Hindi — zero licence cost

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TECHNICAL APPROACH



TECHNOLOGY USED

- ❖ **Data:** xarray · NetCDF · GRIB2 (ecCodes) · NOAA AWS Open Data
- ❖ **Geo:** GeoPandas · Shapely — 641 Census-2011 districts
- ❖ **Machine learning:** LightGBM — regime classifier, correctors, heavy-rain probability
- ❖ **Deep learning:** PyTorch U-Net corrector · CNN classifier (tested)
- ❖ **Verification:** RMSE · ETS · FSS · Brier · AUC · block bootstrap
- ❖ **Backend:** Python 3.12 · FastAPI · Pydantic
- ❖ **Frontend:** React · TypeScript · Leaflet offline maps · Hindi / English
- ❖ **Quality & deploy:** 179 automated tests · Docker · Git

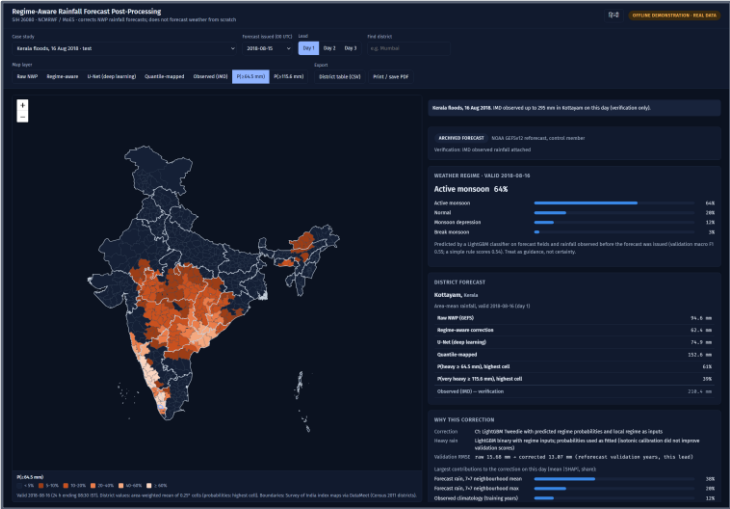


- ✓ Observations never used as model inputs
- ✓ Held-out years scored once, model files hashed

WORKFLOW

- 1. Ingest**
NOAA GFSv12 forecast + IMD 0.25-degree rainfall
- 2. Align & validate**
IMD 08:30 IST rain day · provenance tags
- 3. Identify regime**
LightGBM on 850 hPa wind, MSLP, moisture
- 4. Correct rainfall**
LightGBM C1 · U-Net · quantile mapping
- 5. Heavy-rain probability**
 $P(\geq 64.5 \text{ mm})$ and $P(\geq 115.6 \text{ mm})$
- 6. District forecast**
Area-weighted onto 641 districts
- 7. Serve**
FastAPI → dashboard · CSV · PDF bulletin
- 8. Verify**
Against IMD, with bootstrap confidence intervals

PROTOTYPE — RUNNING



Dashboard · Kerala floods case (test year 2018)

```
> python scripts/run_live.py --latest
--allow-out-of-season
INFO downloaded 2026-09-28 in 16s
INFO 2026-09-28 saved (29s total):
  day-1 regime BREAK (84%),
  verification: pending: IMD has not
  published observations for these days
```

Real terminal output: today's NOAA forecast → district product in 29 s

✓ FEASIBILITY ASSESSMENT

- ❖ **Technical:** built and running end-to-end on real data — 16 training seasons, 29.1 M forecast–observation pairs, 179 automated tests
- ❖ **Operational:** live mode turns today’s forecast into district products in < 1 min; offline dashboard; one-command Docker
- ❖ **Economic:** open data (NOAA, IMD) and open-source stack on a laptop CPU — no GPU, zero licence cost
- ❖ **Social:** English / Hindi interface, district CSV and printable bulletins for disaster managers

18

monsoon seasons (2000–17)

29.1 M

training pairs

641

districts covered

179

automated tests passing

⚠ POTENTIAL CHALLENGES

- ❖ NCMRWF’s NCUM forecast archive is not public
- ❖ Daily operational forecasts differ from the training archive (tree correctors run ~1.2 mm too dry)
- ❖ Rainfall-amount forecasts under-forecast extremes
- ❖ Models are trained for June–September only



🛡 MITIGATION STRATEGIES

- Validated on NOAA GEFS; NCUM plugs into the same adapter and retrains
- U-Net layer is nearly unbiased on live forecasts; retrain on 2021–25 operational seasons
- Heavy-rain probability layer catches extremes (AUC 0.90 on test years)
- Season guard with a visible warning; extend training months

VIABILITY
path to
operations

① Prototype ✓
validated + held-out test

② Live mode ✓
610 real NOAA forecasts

③ NCMRWF data
NCUM / NEPS retrain

④ Daily operations
IMD / NCMRWF dashboards

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IMPACT AND BENEFITS



MEASURED IMPACT — unseen real data, scored once

0.56 → 0.90

heavy-rain detection (ROC AUC), test years 2018–19

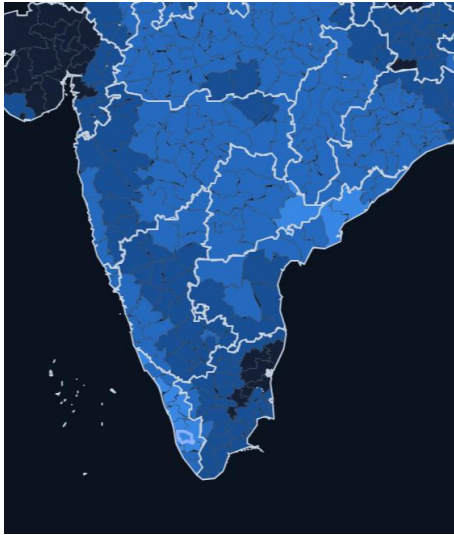
–20%

day-1 error, test years 2018–19 (–12% on 610 live forecasts)

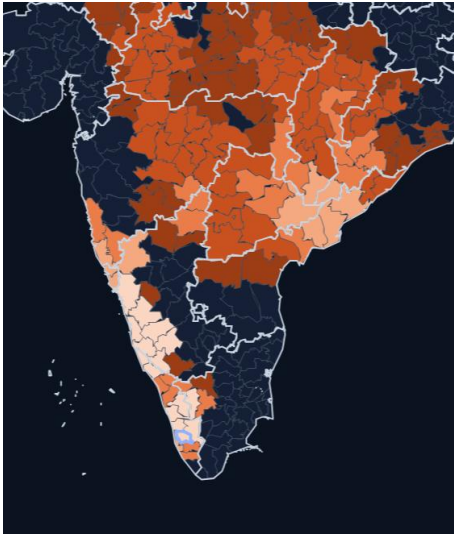
61%

heavy-rain chance flagged for Kottayam before the Kerala floods

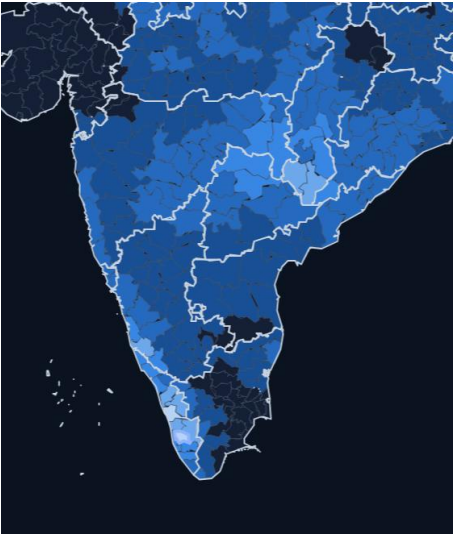
BEFORE · raw forecast



AFTER · our heavy-rain chance



WHAT HAPPENED · IMD



Kerala floods · rain day ending 08:30 IST, 16 Aug 2018 (test year), forecast issued 15 Aug
Kottayam: raw 95 mm · regime-aware 62 · U-Net 75 · observed 210 mm — amounts fall short; the probability layer flags 61%

IMPACTS & BENEFITS

SOCIAL

Earlier district heavy-rain warnings for flood & landslide preparedness

ECONOMIC

Calibrated odds for sowing, irrigation and reservoir decisions

ENVIRONMENTAL

Better water-resource management across monsoon spells

OPERATIONAL

Plugs onto existing NWP output — no new weather model

TECHNOLOGICAL

Explainable AI: each forecast ships its drivers and score

NATIONAL

Indigenous open-data forecasting aid; supports UN SDG 13

Who benefits



IMD / NCMRWF forecasters



NDMA / SDMA disaster mgmt



Farmers & agri planners



Reservoir & water managers



Citizens in flood-prone areas

 KEY FINDINGS — validated on 2016–17, confirmed on held-out data

**Regime-awareness helps — modestly, provably**
Beats the regime-blind model on error in validation and on both held-out sets (test: +0.08 mm, 95% CI 0.03–0.14)

**Deep learning wins on heavy rain**
U-Net heavy-rain ETS 0.155 vs 0.098 (test years) and 0.096 vs 0.070 (610 live forecasts); wetter on average

**Reported where it loses**
In monsoon depressions raw GEFS places heavy rain best; tree correctors run ~1.2 mm dry on live forecasts

#	Title of paper / dataset	Published in (click)	Year	Used for
1	Active and break spells of the Indian summer monsoon — Rajeevan, Gadgil & Bhate	J. Earth Syst. Sci.	2010	Criterion for active / break regime labels
2	0.25° gridded daily rainfall over India — Pai et al.	MAUSAM (IMD)	2014	Ground truth for training and verification
3	GEFSv12 reforecast and operational forecasts	NOAA Open Data on AWS	2000–25	Forecast input: 20 reforecast seasons + live runs
4	Scale-selective verification of rainfall (FSS) — Roberts & Lean	Monthly Weather Review	2008	Spatial skill of heavy-rain placement
5	LightGBM: a highly efficient gradient boosting decision tree — Ke et al.	NeurIPS · LightGBM docs	2017	Regime classifier, correctors, heavy-rain model
6	U-Net: convolutional networks for image segmentation — Ronneberger et al.	MICCAI · arXiv	2015	Whole-map deep-learning corrector
7	Cyclone and monsoon-depression best track	IMD RSMC New Delhi	1982–2025	Monsoon-depression regime labels
8	India district boundaries (Census 2011, Survey of India index maps)	DataMeet	2011	641-district aggregation